Visualization, Customer Data, Intro

MGT 100 Week 1

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Data Visualizations (Viz)

- “The greatest value of a picture is when it forces us to notice what we never expected to see.”

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- Boxplot Inventor John Tukey
- Great visualizations raise new questions
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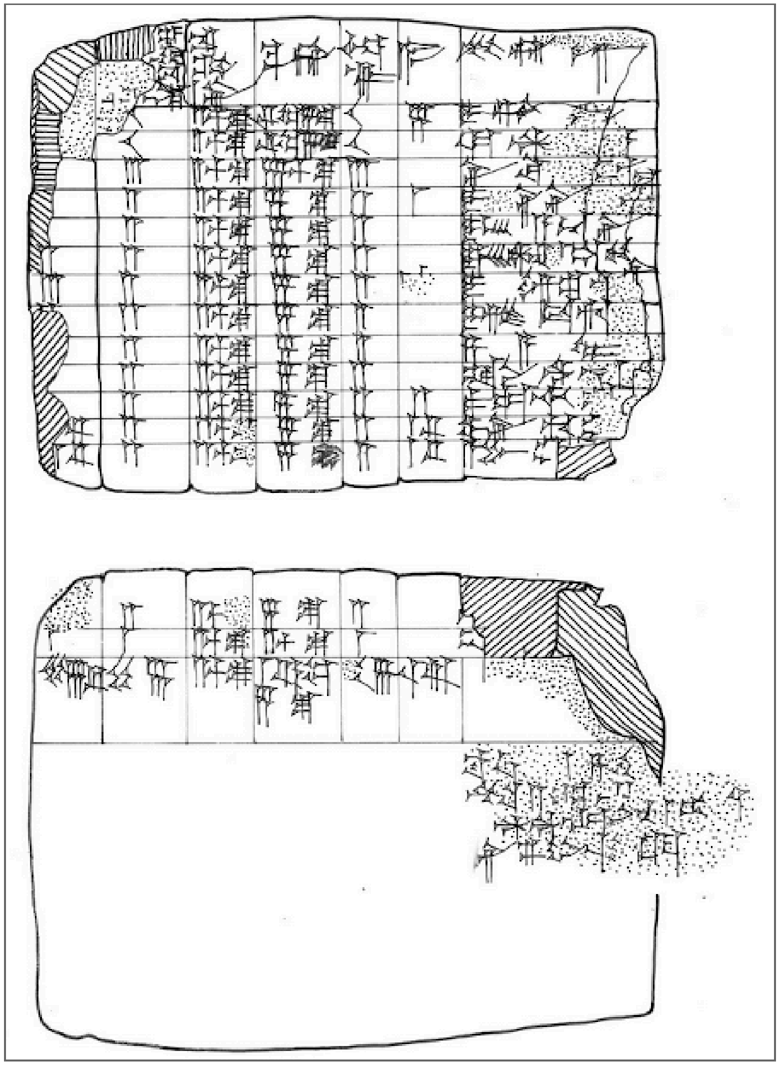
Prehistoric Data Viz

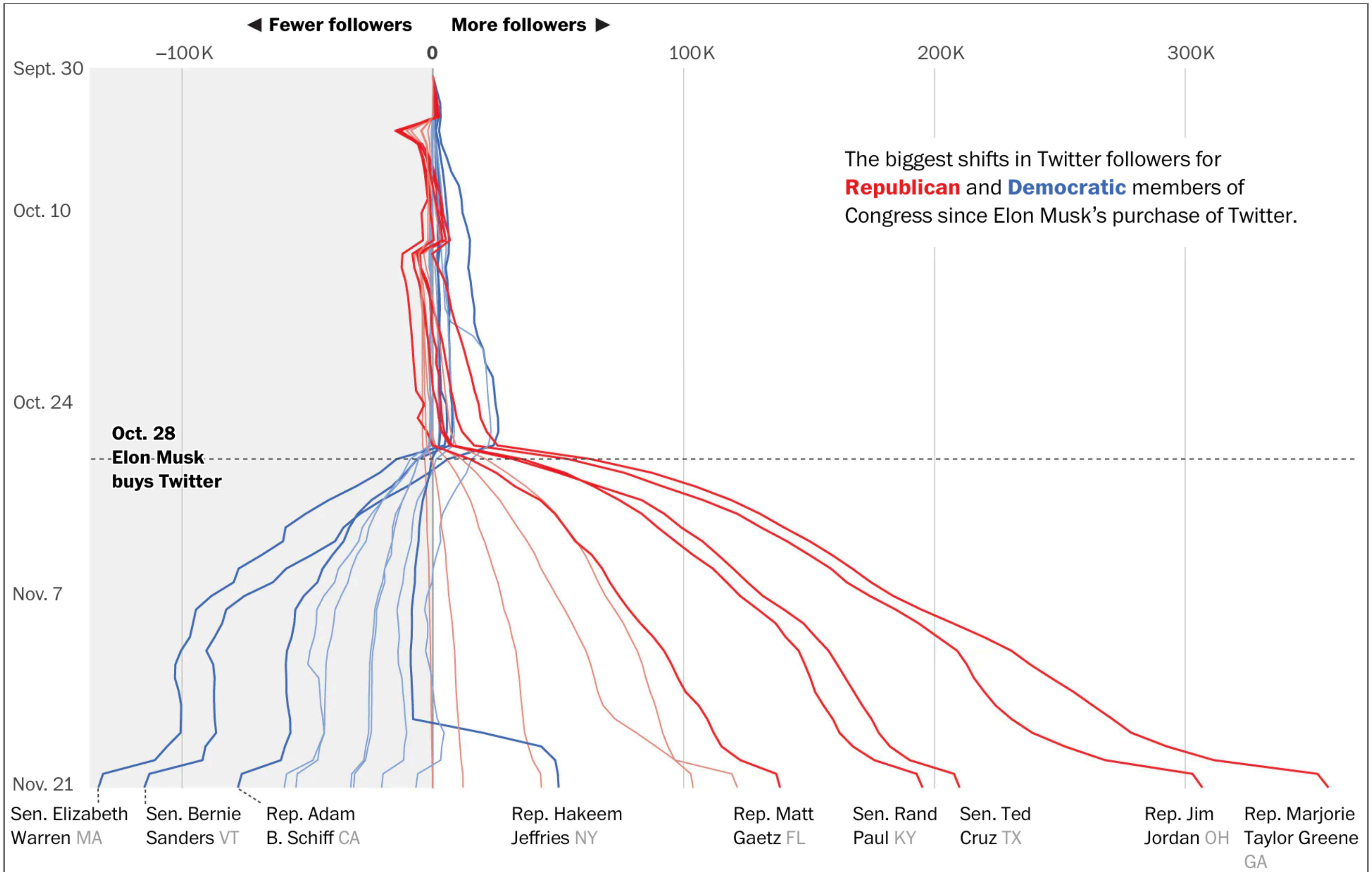


Translated

A	B	C	D	E	F	G
[15]	10	2½ s.	⅓ m. 5 s.	10	5	Igigi
[10]	10	2½ s.	⅓ m. 5 s.	10		Awat-Šamaš
[3]	3	2½ s.	7½ s.	3		Nidittum
[3]	2	2½ s.	5 s.	2	1	The sons of Ili-iddinam
[2]	2	2½ s.	5 s.	2		The sons of Ahuni
[2]	2	2½ s.	5 s.	2		The sons of Sin-šamuh
3	2	2½ s.	5 s.	2	1	The son of Imgurrum
2	2	2½ s.	5 s.	2		Isi-dare
[2]	2	2½ s.	5 s.	2		Zayyalum
[2]	2	2½ s.	5 s.	2		Silli-Ištar
[4]	2	2½ s.	5 s.	2	2	Paqatiya
[3] ⅓	2	2½ s.	5 s.	2	1⅓	Perhum
[3] ⅓	2	2½ s.	5 s.	2	1⅓	Abum-ili
[2]	2	2½ s.	5 s.	2		[*Sin-ašareda*]
[1]	1	2½ s.	[2½ s.]	1		Aplum
57 [⅔]	46	2½ s.	1⅙ m. 5 s.	46	11⅔	
Tašritum (Month VI), [day...]						
Year: the third after Isin was destroyed by the great weapons of An, Enlil and Enki						

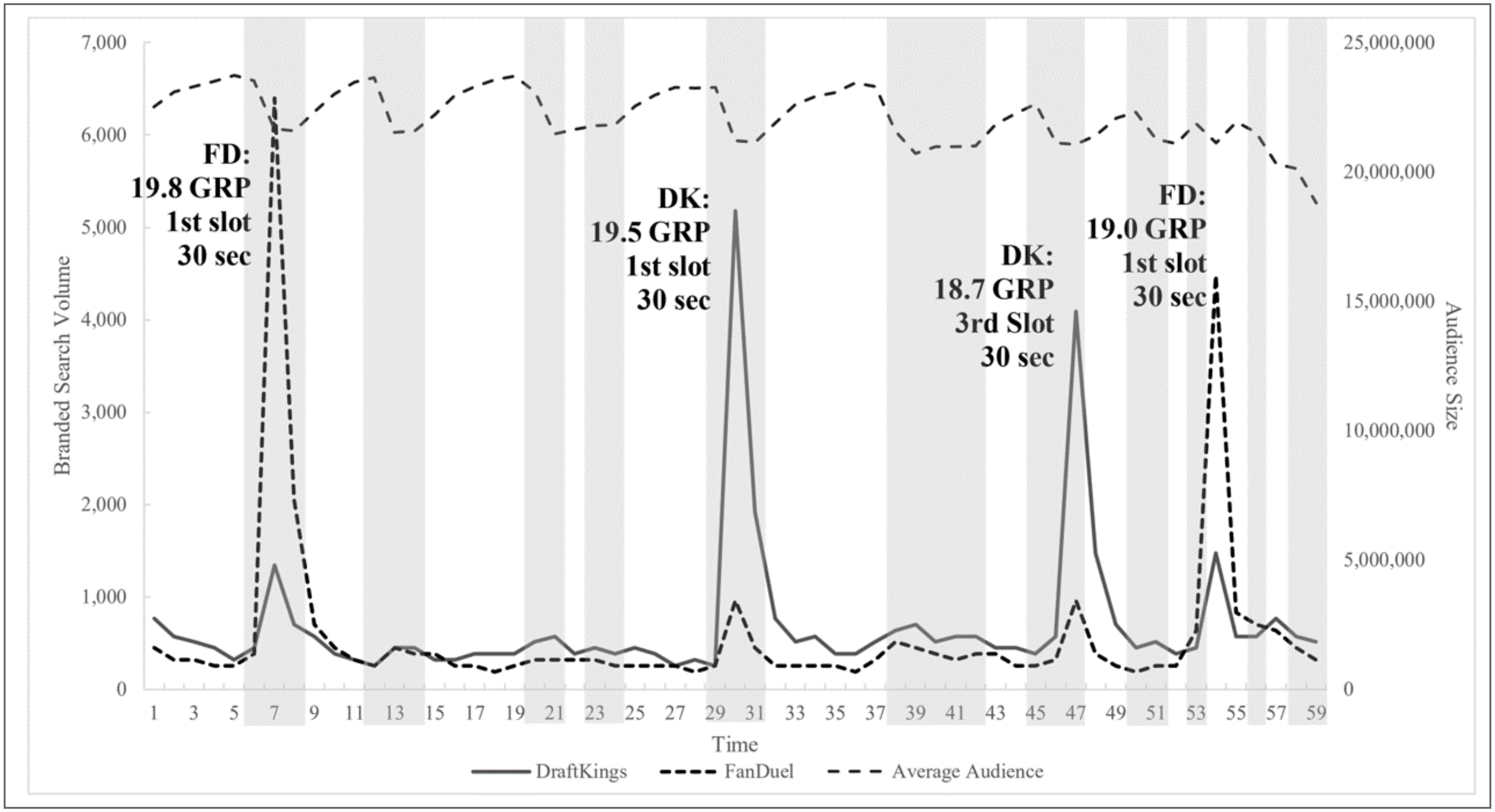
A = "Troops, available assets"
B = "Those who were sent *from* Sin's dyke"
C = "Earth, the (daily) work assignment of 1 man"
D = "Its earth"
E = "Check made"
F = "Arrears"
G = "Its name"

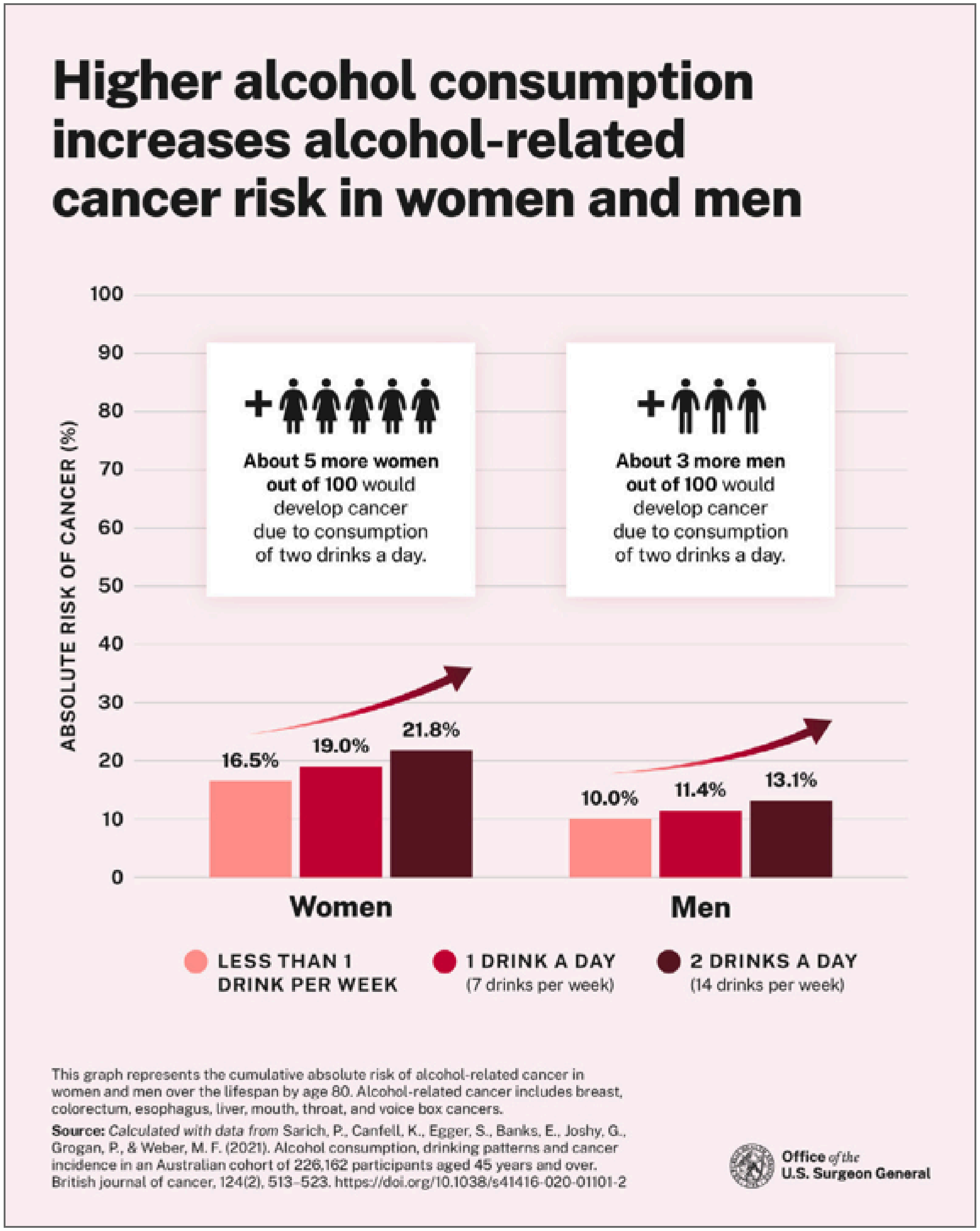
When Elon bought Twitter



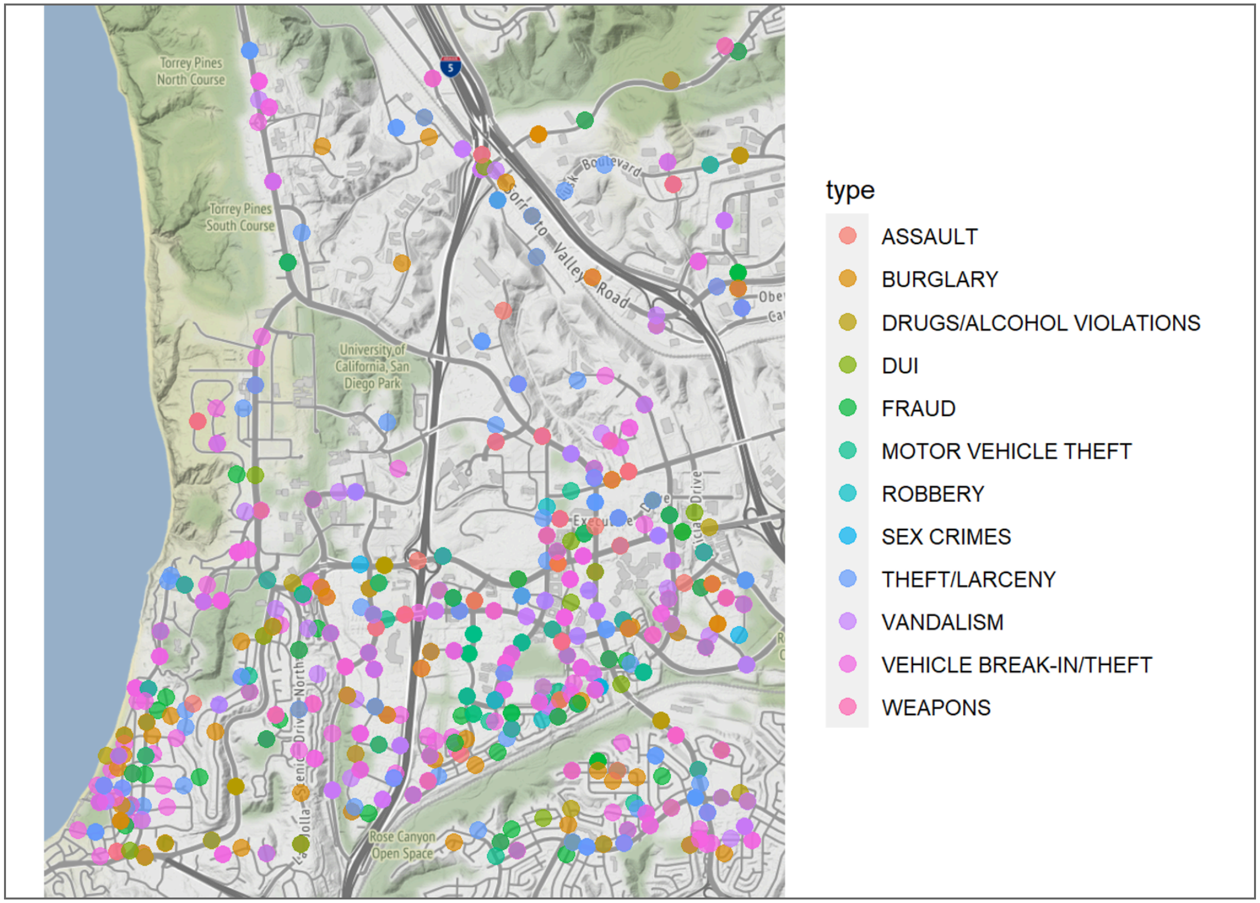
Note: Sanders is an independent who caucuses with Democrats. Top 10 Democrats and Republicans by the difference in follower counts from Oct. 27 to Nov. 21.

DraftKings, FanDuel Searches during an NFL Game, 9-10 P.M.




Higher alcohol consumption increases alcohol-related cancer risk in women and men
ABSOLUTE RISK OF CANCER (%)
100
90
80
70
60
50
40
30
20
10
0
About 5 more women out of 100 would develop cancer due to consumption of two drinks a day.
About 3 more men out of 100 would develop cancer due to consumption of two drinks a day.
16.5%
19.0%
21.8%
10.0%
11.4%
13.1%
Women
Men
LESS THAN 1 DRINK PER WEEK
1 DRINK A DAY (7 drinks per week)
2 DRINKS A DAY (14 drinks per week)
This graph represents the cumulative absolute risk of alcohol-related cancer in women and men over the lifespan by age 80. Alcohol-related cancer includes breast, colorectum, esophagus, liver, mouth, throat, and voice box cancers.
Source: Calculated with data from Sarich, P., Canfell, K., Egger, S., Banks, E., Joshy, G., Grogan, P., & Weber, M. F. (2021). Alcohol consumption, drinking patterns and cancer incidence in an Australian cohort of 226,162 participants aged 45 years and over. British journal of cancer, 124(2), 513–523. https://doi.org/10.1038/s41416-020-01101-2
Office of the U.S. Surgeon General

SDPD Crime Reports Near UCSD



- When is data zero vs. missing? Provenance is key

Data *Provenance*

- describes people, entities, and activities involved in producing, compiling, transforming & sharing data

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- Verifiability requires source & cleaning scripts
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- Crucial in determining quality, reliability, trustworthiness
- Important aspects: Customer consent; Data privacy; Missingness remedies; Contractually permissible uses
- When can you fully trust provenance information?

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- Best to treat provenance as hypotheses to be verified
- Usually, provenance descriptions are missing or imperfect
- Sometimes, provenance descriptions are marketing documents
```

Questions to ask

- Data Origin, Measures, Quality, Verifiability

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-   Where did the data come from?
-   Who collected it, when and why?
-   How is each variable measured? (surveys, APIs, sensors, etc; units)
-   Is there a data dictionary or changelog?
-   How can I verify individual datapoints?
-   Is the data certified by any external party?
```

- Processing & Transformation

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-   What cleaning or manipulation was done?
-   Who performed it and why?
-   Were missing values imputed? How?
-   How were outliers handled?
```

Viz can build trust, understanding

- Viz are *lingua franca* across disciplines
- Eyeballs can interpret pictures quickly

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- Our brains evolved to interpret visual patterns and make predictions
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- Easily replicated -> more easily trusted
- Understandable to managers
- Can detect unknown errors
- Best when they raise deeper questions

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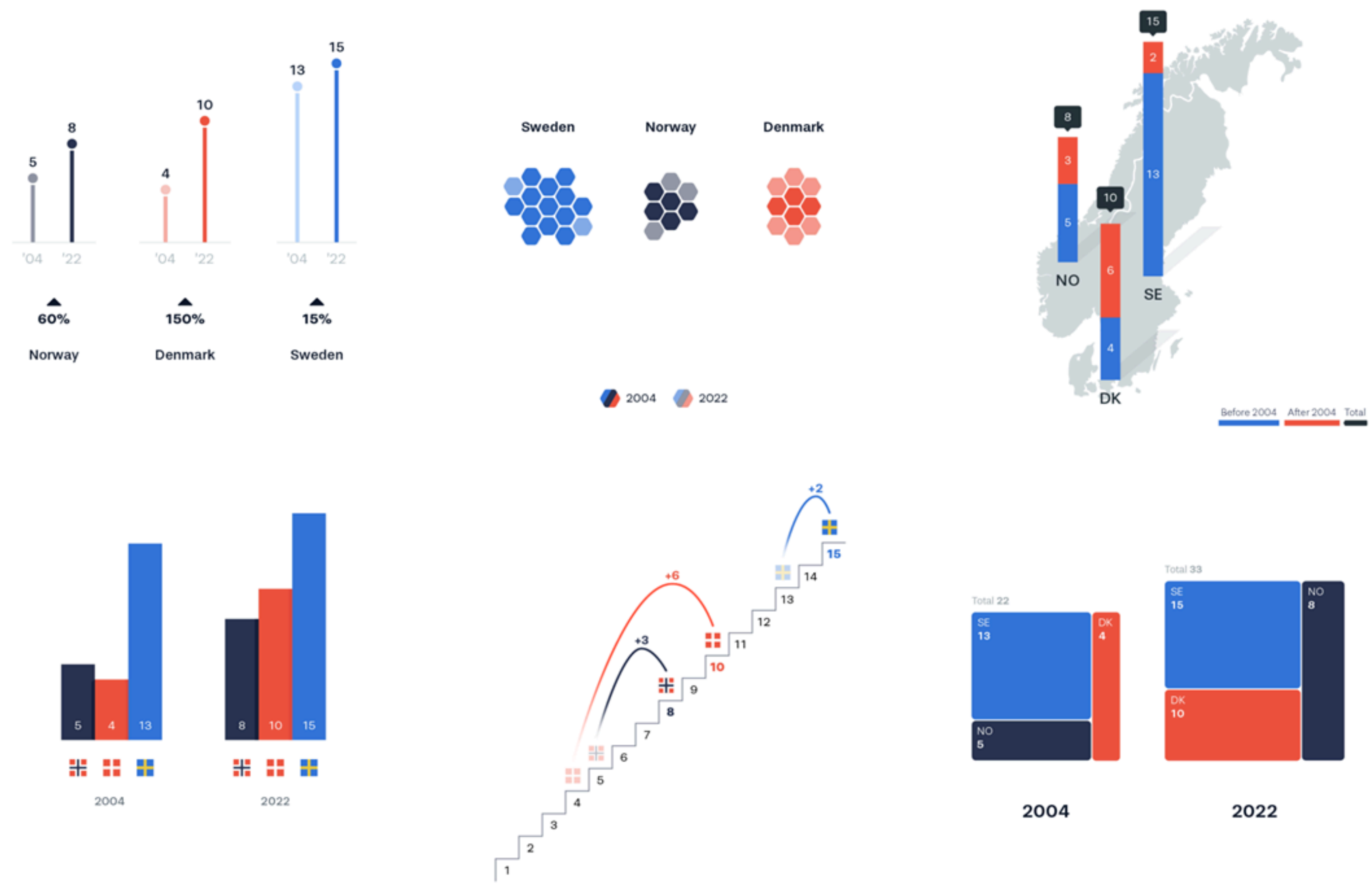
    - Asking the next question indicates trust
```

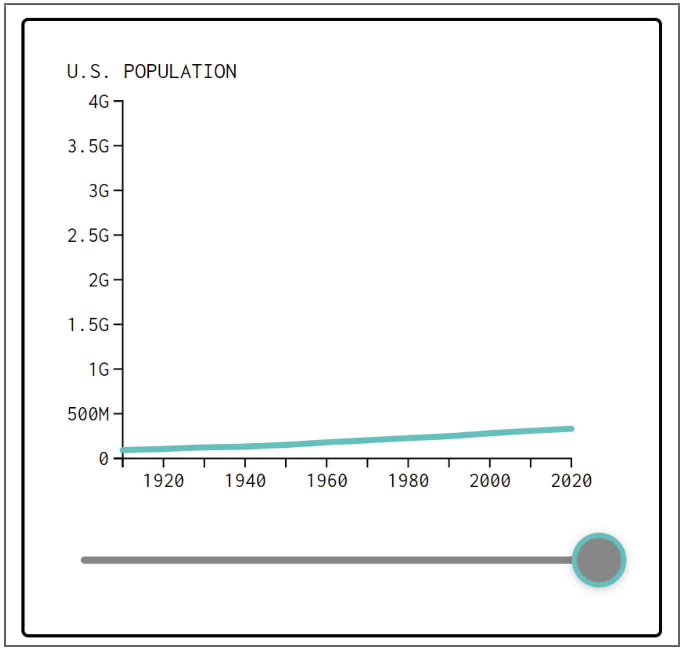
Number of World Heritage Sites

	Norway	Denmark	Sweden
2004	5	4	13
2022	8	10	15



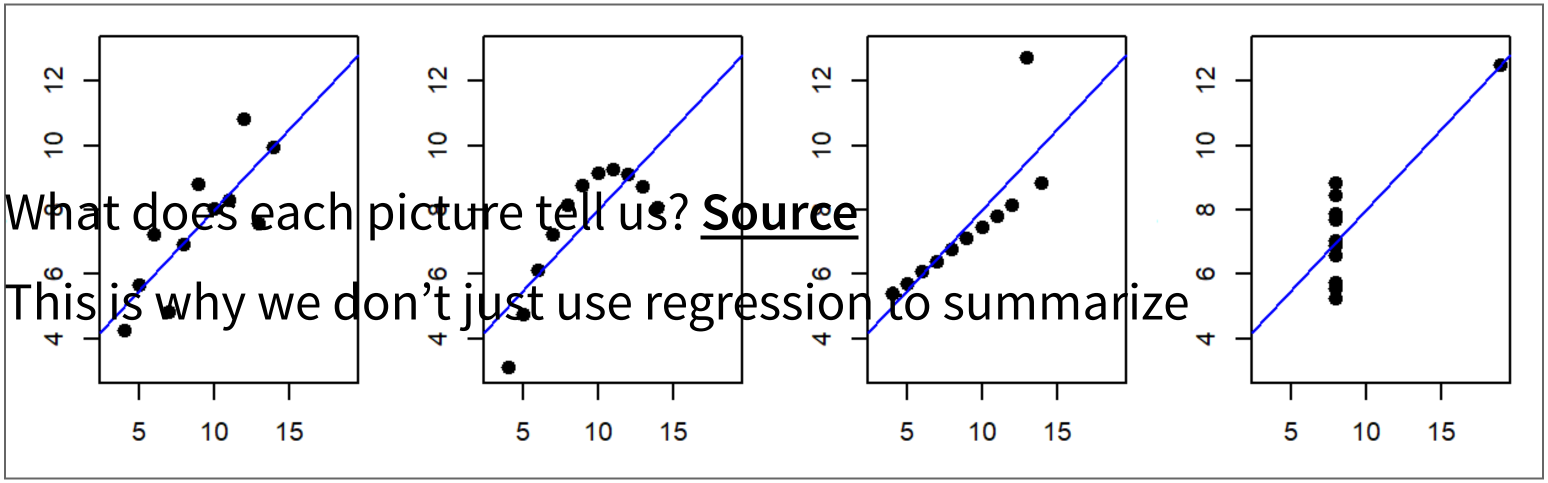
How to Not Mislead with Viz

1. Start axes from 0 and choose scales judiciously
2. Do not add, edit or trim data points
3. Limit discretizations and smoothing



Visualize data before you analyze

- 4 Datasets, Same $\hat{\beta}^{OLS}$, i.e. same $\frac{x'x}{x'y}$



- What does each picture tell us? **Source**
- This is why we don’t just use regression to summarize

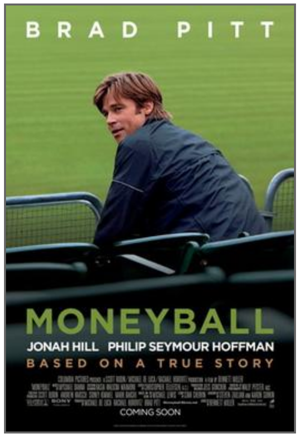
Customer Analytics

Customer

- Receives good or service in exchange for payment (money, time, attention)
- Has agency: Can say “no”
- “The purpose of business is to create and keep a customer.”
  -Drucker

Analytics

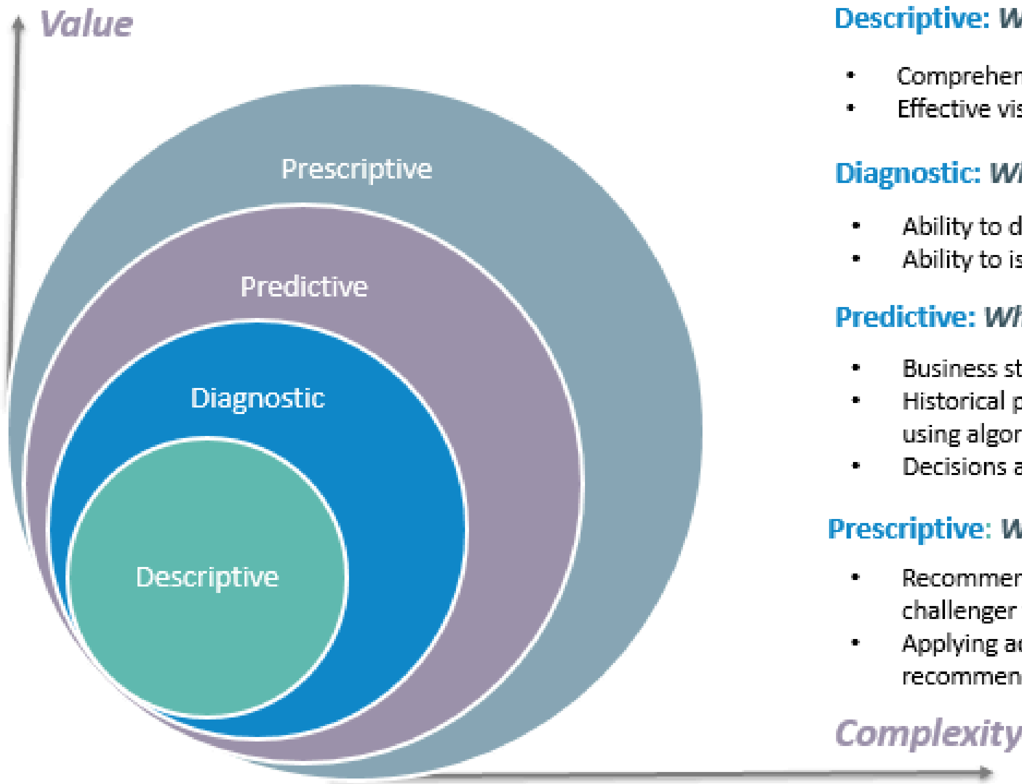
- Using data to improve decisions
- Started by Charles Taylor in the 1880s
- Popularized by *Moneyball* (2011)
- Measurement, Heuristics, Graphics, Models, Predictions, Automation, …
- Can be deceptively difficult



First Law of Customer Analytics

- No Customers, No Business
- No Customers ->
  No Revenue ->
  No Profit ->
  No Business

4 types of Data Analytics



What is the data telling you?

Descriptive: *What's happening in my business?*

- Comprehensive, accurate and live data
- Effective visualisation

Diagnostic: *Why is it happening?*

- Ability to drill down to the root-cause
- Ability to isolate all confounding information

Predictive: *What's likely to happen?*

- Business strategies have remained fairly consistent over time
- Historical patterns being used to predict specific outcomes using algorithms
- Decisions are automated using algorithms and technology

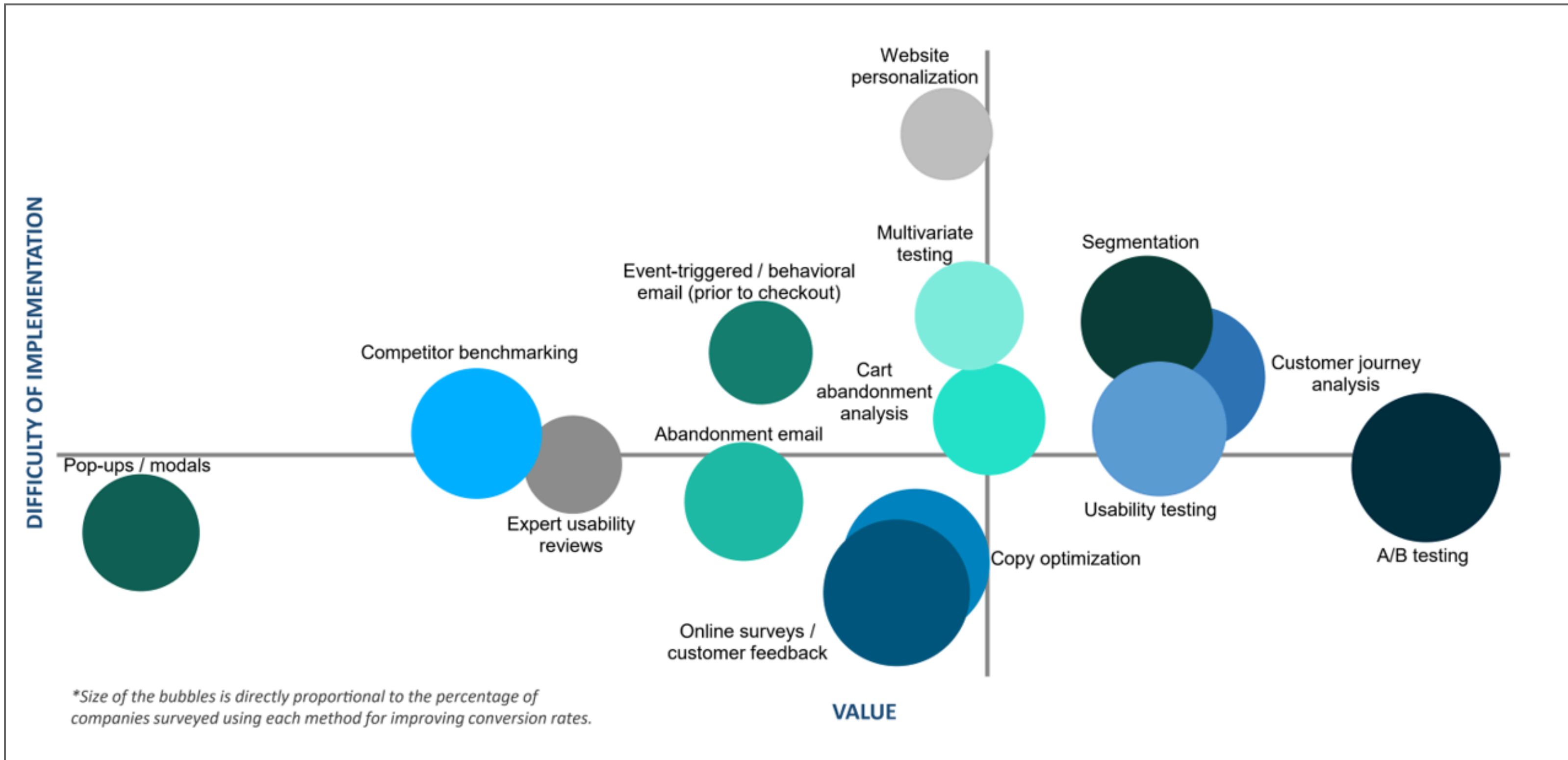
Prescriptive: *What do I need to do?*

- Recommended actions and strategies based on champion / challenger testing strategy outcomes
- Applying advanced analytical techniques to make specific recommendations

Principa
www.principa.co.za

E-commerce Funnel

E-commerce Analytics



- ...ers?
- ... business intelligence, data science

Conversion Rate Optimization Report

Customer Data

Customer Data

- Customer data show how customers and consumers learn, feel, behave and use products and services
- Customer data increasingly drives marketing, but implementation varies widely

How can we use customer data?

- Customer relationship management: Acquire, develop, retain and “fire” customers
- Marketing mix: Improve product offerings, prices, promotion, distribution
- Understand customer heterogeneity for targeting, personalization, recommendations, product development…
- Privacy and security, e.g. misuse, theft, regulatory compliance

Example: Nielsen

CONSUMER PANEL DATA

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      - The Consumer Panel Data include longitudinal data
 beginning in 2004 with annual updates. These data track a panel of
 40,000–60,000 US households and their purchases of fast-moving
 consumer goods from a wide range of retail outlets across all US
 markets.
```

RETAIL SCANNER DATA

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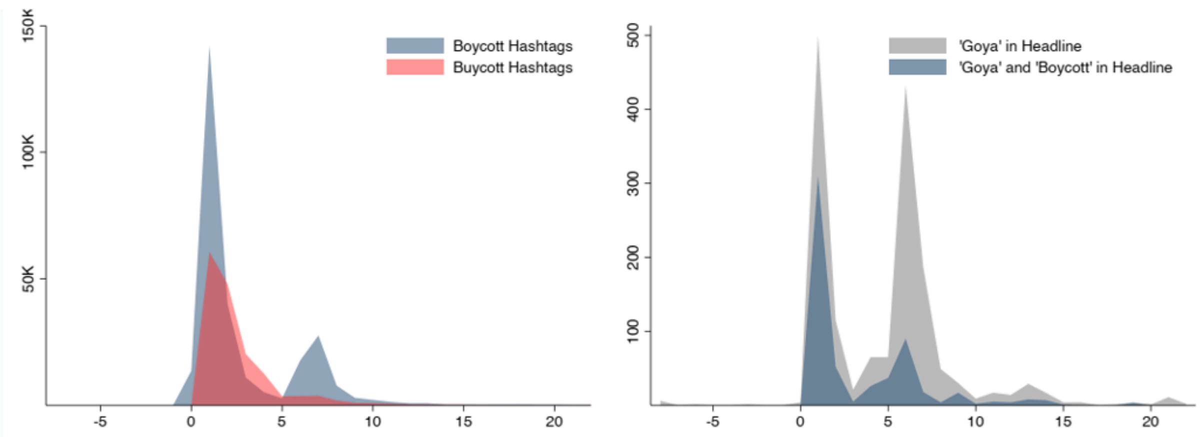
      - Retail Scanner Data consist of weekly pricing, volume, and
 store environment information generated by point-of-sale systems
 from more than 90 participating retail chains across all US
 markets. Data begin in 2006 and include annual updates.
```

Figure 1: Images of Tweets Triggering Goya Boycott and Buycott

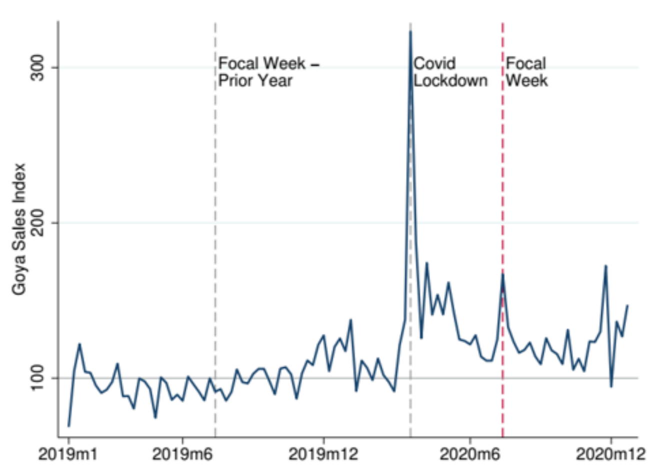
Figure 2: Goya Related Daily Tweets and Media Coverage in July 2020

(a) Daily Tweets **(b)** Daily Media Articles



Notes: Day 0 is July 9th, 2020. Panel (a) depicts the number of daily tweets in July 2020 that use boycott- or buycott-related hashtags. Web Appendix A.3 details the classification procedure and Table A1 lists the hashtags. Panel (b) plots the number of daily media articles that include 'Goya' and 'Boycott' in the headline. Only 18 articles included buycott related keywords so they are not depicted. In both panels, the second spike is triggered by President Trump and his daughter's tweets in support of Goya.

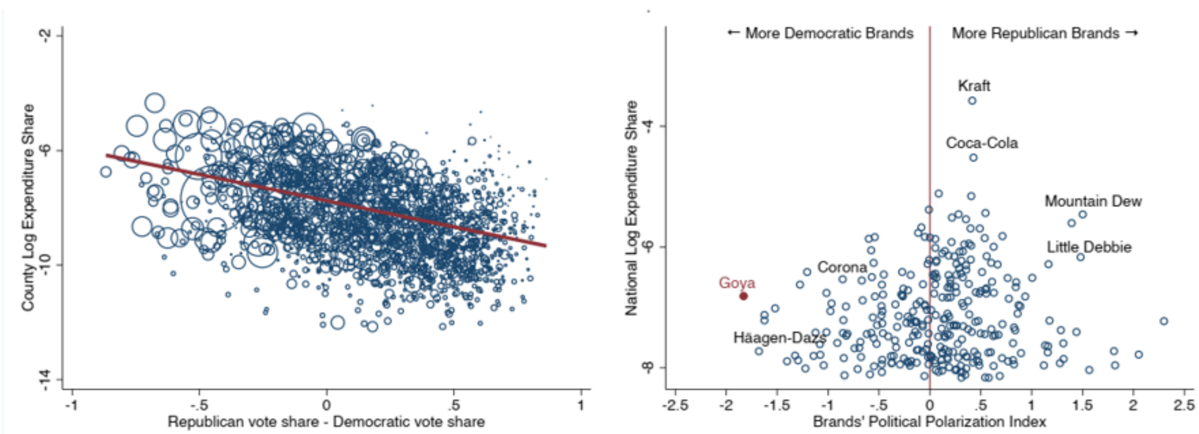
Figure 3: Total Weekly Spending Index on Goya Products



Notes: This figure depicts weekly spending index on Goya products from 2019-2020. The pre-pandemic sales average is normalized to 100. The red vertical dashed line indicates the focal week of the scandal (July 9-15). The spike and subsequent dip in November 2020 is due to bunching in purchases around the Thanksgiving holiday. Please see Appendix B.8 for a replication of the sales trend using Nielsen RMS data.

Figure 5: Polarization in Brand Popularity: Goya vs Other Top Brands

(a) Goya Market Share and Vote Margin **(b)** Polarization of Goya and Other Top Brands

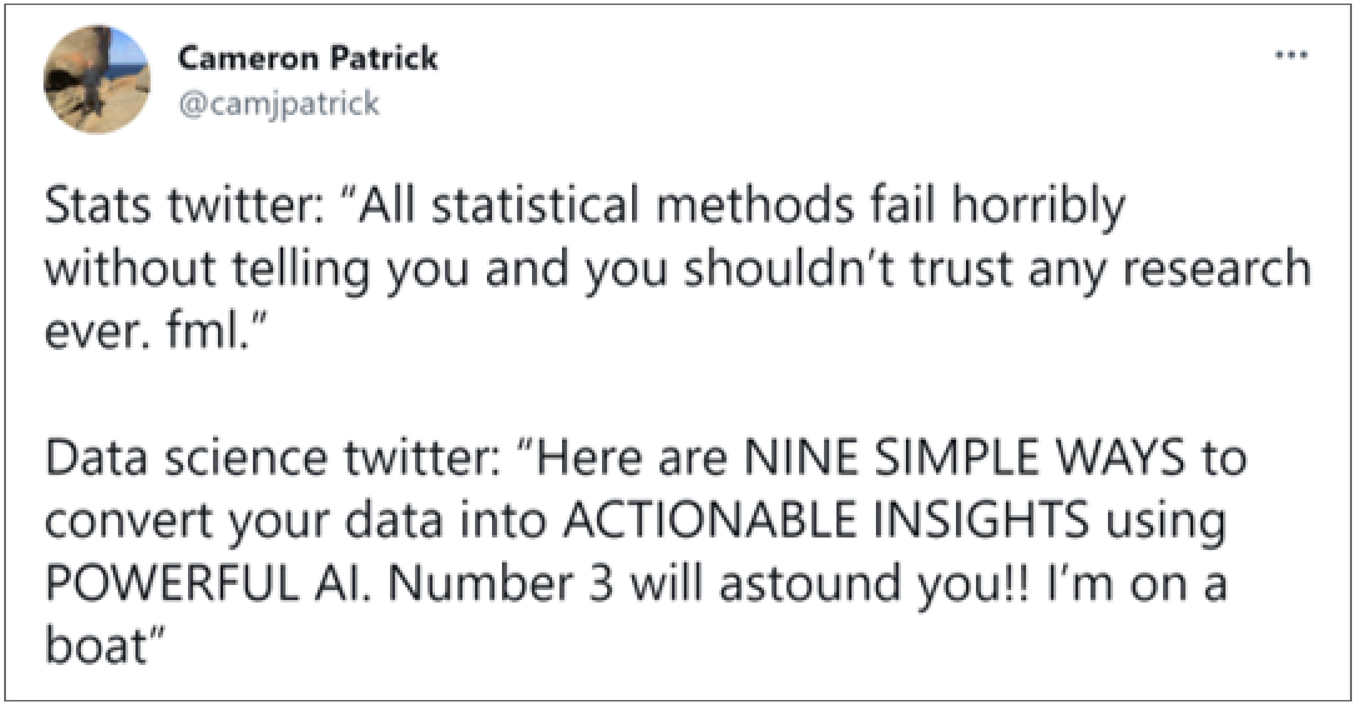


Notes: Panel (a) depicts the relationship between Goya's log expenditure share in a given county in 2019 and the vote margin in that county. The size of the bubble is proportional to the number of panelists in a given county. Panel (b) plots the estimated political polarization index for each of the top 301 CPG brands. The index quantifies the extent to which the market share of a brand varies with political preferences. Goya's national expenditure share is 0.11% (the log of its national expenditure share is −6.81). See Web Appendix A.1 for more detail.

Customer Data: Guiding Principle

- Start simple. Complexify slowly. Why?
- Never assume data are correct, clean, complete or as described.

```
- It's impossible to certify an absence of problems
- Most commercial datasets have issues
- Issues often detected months after project starts
- ~70-90% of data scientist time spent checking and cleaning data
- Credibility is hard to gain, easy to lose
```


Cameron Patrick
@camjpatrick
Stats twitter: “All statistical methods fail horribly without telling you and you shouldn’t trust any research ever. fml.”
Data science twitter: “Here are NINE SIMPLE WAYS to convert your data into ACTIONABLE INSIGHTS using POWERFUL AI. Number 3 will astound you!! I’m on a boat”

Using Customer Data for Customer Analytics

How do we “do” customer analytics?

- Decide objectives & outcome metrics
- Collect, wrangle, clean & verify relevant data
- Analyze data
- Communicate analyses and recommendations
- Make decisions
- Implement data-driven decisions
- Retrospectively evaluate and improve
- Repeat
- …Once you have a stable process, automate carefully & monitor

Challenges: Executives

- May be territorial, or incentivized to be
- May worry that analytics will constrain or replace them
- May think data == magic
- May prefer hunches or misunderstand uncertainty

Challenges: Analysts

- Expensive
- Hard to find
- Not always current
- Not always interested in business

Challenges: Cultural

- Do analytics make or justify decisions?
- High- or low-trust environment? Tolerance for uncertainty?
- Do messengers get rewarded or shot?
- Are data available and integrated?
- Do teams work together or compete?

Signs of a great analytics org

- C-level champion(s), i.e. C{D,E,A,F,M,O}O
- Centralized team regulates data, arch., standards & tools
- Decentralized analysts collaborate with execs
- Analytics career tracks are well established
- Careful in-housing/outsourcing decisions about analytics
- Good examples?

Analytics truisms

- Analytics matters more in B2C than B2B (why?)
- Selection effects are usually large

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treatment effects are usually small
```

- Demographics don't predict behavior very well
- Agencies lie about data sometimes
- You have limited credibility. You may only get a few strikes
- "If it's written in LaTeX, it's probably correct"

Our Class

Course Design Principles

- Survey the field with pointers for deeper learning
- Focus on understanding:
  “Skillful LLM use will run laps around unskilled LLM use”
- Free materials
- Communication

```
1. Website for syllabus, slides, scripts, data, readings
2. Canvas for groups, deliverables & grades
3. Piazza for all asynchronous interaction. No email or canvas messages
4. After class, break or office hours for live discussions
```

- **Course site & schedule**

Tips to get an A

- Read the syllabus carefully
- Contribute in class as suggested in the syllabus
- Budget 5-10 hours/week
- Between classes:

```
1. Do homework solo
2. Check homework with group, resolve differences
3. Monitor Piazza
4. Read for the next class
5. Fix a time/location, repeat 8 times
```

Studying and Integrity

- We assign attending students to study groups in week 2
- It is OK to share homework scripts and answers

```
- We do not collect or assess homeworks
- Exam questions will test your familiarity and understanding of homework answers
```

- We encourage you to use Gen AI; we use it too

```
- Be advised, you may get what you pay for
```

- Please write down your intentions for this class. How will you measure your effort?
- Midterms and final will be open-paper, closed-device. In-person attendance required.
- How do survey courses differ from other classes? | **Why R?**

Vocab

Common language helps communication

Customer level

- Core need: identifiable problem a customer wants to solve. Could be functional, emotional, social, profit-motivated, etc. Related: desire, want, pain point
- Core benefit: Customer’s desired outcome of a purchase. E.g., commuters need to get to school, not necessarily cars
- Consumer: Entity that experiences the core benefit
- Customer: Entity that purchases and pays

Product level

- Product/service/experience: Distinct offering that provides the core benefit
- Features: Aspects of a product that provide additional tangible or intangible benefits
- Contribution margin: Price — marginal cost
- Competitor: Any paid or free alternative that addresses the core need. E.g., commute by bike, walk, bus, trolley, Uber, scooter, skateboard; work from home

Market level

- Market: A group of potential customers with the same core need
- Segment: Distinct subgroup of similar customers
- Targeting: Which segment(s) a firm tries to serve
- Positioning: Specification of product features to suit targeted segments
- Marketing: Practice of meeting customer needs profitably **How to be good at marketing**

Legacy Terms

- 3/4/5 C’s: Customer, Competitor, Company; Context; Complementors
- 4/…/10 P’s:
  Price, Product, Promotion, Place AKA distribution

  Client

Coding & Script

Coding errors

- “The good news about computers is that they do what you tell them to do. The bad news is that they do what you tell them to do.”
- Conjecture:
  (debugging difficulty) is exponential in (lines of code)
- We can code fast or slow

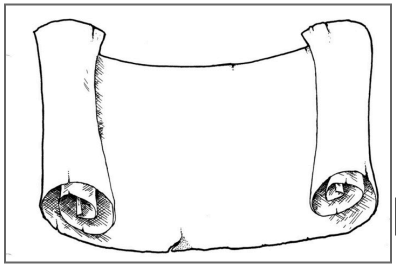
Coding Habits

- Good habit: Test chunks as you code
- Test = Check that output matches expectation
- “Go slow to go fast”

Pipe |>

- y <- f(g(x)) is the same as
- y <- x |>
g |>
f
- Why?
- Old pipe was %>%

Today's script

- **Install/update R and Rstudio**
- **Posit.Cloud**
- Data Import/export
- Data manipulation, summarization
- 5 verbs: Summarize, select, filter, arrange, mutate, group_by
- Univariate statistics
- Univariate plots
- Bivariate statistics
- Bivariate plots
- he script, run it, see where it breaks
- Digression: Production code vs prototype code

Wrapping up

Recap

- Data Viz are best way to start in customer analytics
- Customer analytics : Using customer data to improve decisions
- Marketing : Meeting customer needs profitably
- Analytics types:
  Descriptive, Diagnostic, Predictive, Prescriptive
- Summarize, select, filter, arrange, mutate, group_by

Going further

- Marketing Analytics for Data-Rich Environments
- GGplot2 YT video
- *R for Data Science (2e)*
- *R for Marketing Research and Analytics*
- GGplot2 Book
- *Advanced R*